The product condition without cutoff on unweighted Abelian Cayley graphs

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Abstract

We construct finite simple connected undirected Cayley graphs of Abelian groups whose rate-one simple random walks satisfy the spectral product condition but do not exhibit total-variation cutoff. The groups are products of a hypercube group and a cyclic group. Uniform choice from an explicit generating set induces an exact rare-reset mechanism on the cube factor, while the auxiliary factor mixes on a negligible time scale. We compute the complete character spectrum and the limiting total-variation profile, obtaining an explicit separation between two fixed-threshold mixing times. The construction answers the unrestricted formulation of the Abelian Cayley question in Salez's survey and the corresponding bounded-step nilpotent-group conjecture of Hermon and Olesker-Taylor. Degrees grow with the graph size; bounded-degree and random-generator formulations are unaffected. The reset mechanism is classical. The proposed contribution is its explicit unweighted realization and the resulting source-question counterexample; novelty remains subject to specialist prior-art review.

1 Statement and scope

For a finite connected undirected regular graph let P be its simple-random-walk transition matrix and let $e^{t(P-I)}$ be its rate-one continuous-time semigroup. Write $v(t)$ for the worst-start total-variation distance to stationarity and

$$t(\varepsilon) = \inf\{t \geq 0 : v(t) \leq \varepsilon\}, \qquad 0 < \varepsilon < 1.$$

The spectral gap is $\gamma = 1 - \lambda_2(P)$. A sequence has cutoff if $t(\varepsilon)/t(1-\varepsilon) \to 1$ for every $0 < \varepsilon < 1/2$. The product condition is $\gamma t(1/4) \to \infty$.

For integers $d \geq 8$, set

$$N_d = 2^d, \qquad m_d = 1 + \left\lceil \frac{2^d \log d}{4} \right\rceil, \qquad H_d = (\mathbb{Z}/2\mathbb{Z})^d, \quad K_d = \mathbb{Z}/m_d\mathbb{Z}. \tag{1}$$

Let $e_1, \ldots, e_d$ be the standard generators of H_d. In $G_d = H_d \times K_d$ choose the generating set

$$S_d = \{(e_i, k) : 1 \leq i \leq d,\ k \neq 0\} \ \cup\ \{(x, 0) : x \in H_d \setminus \{0\}\}. \tag{2}$$

All generators have the same transition probability. Put

$$D_d = d(m_d - 1) + N_d - 1, \quad \alpha_d = \frac{d(m_d-1)}{D_d}, \quad \theta_d = \frac{N_d - 1}{D_d}, \quad \beta_d = \frac{N_d}{D_d}, \quad T_d = \frac{d \log d}{4\alpha_d}. \tag{3}$$

The rates θ_d and β_d are different: the former counts actual global-type jumps; the latter is the equivalent reset rate in the projected generator.

Theorem 1.1. *The graph* $\mathrm{Cay}(G_d, S_d)$ *is finite, simple, connected, undirected and vertex-transitive, with degree* D_d*. Its rate-one simple random walk satisfies, for every fixed* $s > 0$ *with* $s \neq 1$*,*

$$\lim_{d\to\infty} v_d(sT_d) = \begin{cases} e^{-s}, & 0 < s < 1, \\ 0, & s > 1. \end{cases} \tag{4}$$

For all sufficiently large d *its spectral gap is*

$$\gamma_d = \frac{2(m_d - 1) + N_d}{D_d} = \frac{2\alpha_d}{d} + \beta_d \sim \frac{2}{d}. \tag{5}$$

For every fixed $\varepsilon \in (0, 1)$*,*

$$\frac{t_d(\varepsilon)}{T_d} \longrightarrow \min\{1, \log(1/\varepsilon)\}. \tag{6}$$

In particular,

$$\gamma_d t_d(1/4) \sim \tfrac{1}{2}\log d \to \infty, \qquad \frac{t_d(3/4)}{t_d(1/4)} \longrightarrow \log(4/3) < 1.$$

Thus the product condition holds and cutoff fails.

Salez's Problem 2.4 [1] concerns uniform increments from a generating set and then discusses weighted increments separately. The conjecture in [2, Section 7.4, Conjecture 1] quantifies over sequences of generating subsets of bounded-step nilpotent groups. The present groups have nilpotency class one. These precise unrestricted formulations include (2); neither imposes a degree bound in its displayed statement. Our conclusion does not address uniformly bounded degree, or a high-probability statement for uniformly sampled generators.

2 The graph and the exact projected walk

The two families in (2) are disjoint, since their second coordinates are respectively nonzero and zero. Neither contains the identity. Both are inverse closed: every cube element is its own negative, and $k \neq 0$ implies $-k \neq 0$ in K_d. Thus the Cayley graph is simple and undirected, with the degree asserted in (3). The second family contains each $(e_i, 0)$. Combining (e_1, k) with $(e_1, 0)$ gives $(0, k)$; hence the generators span the full product and the graph is connected. Translations are graph automorphisms, so the graph is vertex-transitive and its stationary distribution is uniform.

Split the rate-one jump process into the two generator types. Independent Poisson thinning gives local-type jumps at rate α_d and global-type jumps at rate θ_d. On a local-type jump the cube flips a uniformly selected coordinate, and independently the auxiliary coordinate adds a uniform nonzero element of K_d. On a global-type jump the cube adds a uniform nonzero element of H_d and the auxiliary coordinate stays put.

Let L_d be the generator of rate-one simple random walk on the cube, and Π_d the operator replacing a distribution on the cube by the uniform law. The kernel of a uniform nonzero cube increment is

$$Q_d = \frac{N_d\Pi_d - I}{N_d - 1}.$$

Consequently the projected generator equals

$$\alpha_d L_d + \theta_d(Q_d - I) = \alpha_d L_d + \beta_d(\Pi_d - I). \tag{7}$$

The cube semigroup commutes with Π_d, which is a projection. Since

$$e^{t\beta_d(\Pi_d - I)} = e^{-\beta_d t} I + (1 - e^{-\beta_d t})\Pi_d,$$

the cube law from zero is the mixture of its ordinary law at time $\alpha_d t$, with weight $e^{-\beta_d t}$, and uniform law with the remaining weight. If $c_d(t)$ is the ordinary rate-one cube distance and $q_d(t)$ is the projected distance, then

$$q_d(t) = e^{-\beta_d t} c_d(\alpha_d t). \tag{8}$$

This is an exact identity, not an approximation replacing a nonzero increment by a uniform reset.

3 Controlling the complete product law

Projection contracts total variation, so $q_d(t) \le v_d(t)$. We now prove a matching approximation from above; convergence of a projection alone would be insufficient.

Lemma 3.1. *For every $t \ge 0$,*

$$0 \le v_d(t) - q_d(t) \le e^{-\alpha_d t} + \frac{1}{m_d}. \tag{9}$$

Proof. Let M_t count the local-type jumps, so M_t is Poisson with mean $\alpha_d t$. Condition on the two jump processes, their times, and all cube increments. The auxiliary coordinate is then a sum of M_t independent uniform nonzero increments. These remain independent under this conditioning, by the independent selections in the definition of each local-type jump.

The distance between the uniform law on $K_d \setminus \{0\}$ and uniform law on K_d is exactly $1/m_d$: half the sum of the absolute mass differences is

$$\tfrac{1}{2}\left(\frac{1}{m_d} + (m_d - 1)\left|\frac{1}{m_d - 1} - \frac{1}{m_d}\right|\right) = \frac{1}{m_d}.$$

Convolution by a probability law contracts total variation and preserves the uniform law. Conditional on $M_t \ge 1$, the auxiliary sum therefore has distance at most $1/m_d$ from uniform. On $M_t = 0$, bound the distance by one. Couple to a uniform auxiliary coordinate for each conditional law. The latter has uniform conditional distribution given the entire conditioning information, hence is independent of the cube endpoint. It follows that the joint law is within $e^{-\alpha_d t} + 1/m_d$ of the product of its projected law and an independent uniform coordinate. The latter product has distance exactly $q_d(t)$ from uniform law on G_d. The triangle inequality proves the upper bound. □

4 Cube cutoff and the limiting profile

We include a proof of the only cube asymptotics used. Under rate-one cube walk, independent coordinates flip at rate $1/d$. Their sign variables have means $e^{-2t/d}$. Product structure gives

$$\chi^2(\mathcal{L}(X_t), \mathrm{Unif}(H_d)) = (1 + e^{-4t/d})^d - 1. \tag{10}$$

For $t = sd\log d/4$ and $s > 1$, the right side tends to zero, since $de^{-4t/d} = d^{1-s} \to 0$. Cauchy–Schwarz gives $c_d(t) \le \frac{1}{2}\sqrt{\chi^2}$, proving convergence to zero.

For $0 < s < 1$, let F be the sum of the coordinate signs. From zero its mean is $a_d = d^{1-s/2}$ and its variance is at most d; under stationarity its mean is zero and variance d. Chebyshev applied to the event $\{F > a_d/2\}$ under both laws gives

$$c_d(sd\log d/4) \ge 1 - 8d^{s-1} \longrightarrow 1.$$

No independence between different times is needed.

The rounding in (1) yields

$$\beta_d T_d = \frac{2^d \log d}{4(m_d - 1)} \longrightarrow 1. \tag{11}$$

Also $\alpha_d \to 1$, $m_d \to \infty$, and $T_d \to \infty$. At time sT_d, the error in (9) tends to zero for each fixed $s > 0$, since $\alpha_d s T_d = sd \log d/4 \to \infty$. Equations (8) and (11), together with the two cube limits, prove (4).

5 The complete spectrum

Characters of G_d are indexed by a subset $A \subseteq \{1, \dots, d\}$ and a character ψ of K_d. Set $a = |A|$. The cube character takes value $(-1)^{\mathbf{1}_{\{i \in A\}}}$ at e_i, so its sum over the basis vectors is $d - 2a$. Character orthogonality gives

$$c_\psi = \sum_{k \neq 0} \psi(k) = \begin{cases} m_d - 1, & \psi = 1, \\ -1, & \psi \neq 1, \end{cases} \qquad b_A = \sum_{x \neq 0} \chi_A(x) = \begin{cases} N_d - 1, & A = \varnothing, \\ -1, & A \neq \varnothing. \end{cases}$$

The eigenvalue of the uniform-generator transition matrix is therefore

$$\lambda_{A,\psi} = \frac{(d - 2a)c_\psi + b_A}{D_d}. \tag{12}$$

These characters form an orthogonal basis, so the formula covers the entire spectrum. In particular, there is no unexamined auxiliary bottleneck. The three nonconstant sectors give the generator gaps

$$\begin{aligned} &\frac{2a(m_d - 1) + N_d}{D_d}, && a \geq 1,\ \psi = 1; \\ &\frac{dm_d}{D_d}, && a = 0,\ \psi \neq 1; \\ &\frac{D_d + d - 2a + 1}{D_d}, && a \geq 1,\ \psi \neq 1. \end{aligned}$$

The smallest gap in the first sector has $a = 1$. Its value tends to zero and is asymptotic to $2/d$, whereas the second-sector value tends to one and the infimum of the third-sector values tends to one. Thus the first cube mode is the slowest for all sufficiently large d. This proves (5). More explicitly, $D_d \sim d2^d \log d/4$, $\alpha_d \to 1$ and $\beta_d \sim 4/(d \log d)$, giving the asserted asymptotic.

6 Mixing times and the failure of cutoff

The function $v_d(t)$ is nonincreasing, since the semigroup contracts total variation. For $\varepsilon > e^{-1}$, the profile in (4) crosses level ε at the interior point $s_\varepsilon = \log(1/\varepsilon) \in (0, 1)$. For any sufficiently small $\delta > 0$, the limiting distance at $(s_\varepsilon - \delta)T_d$ is strictly above ε and at $(s_\varepsilon + \delta)T_d$ strictly below it. Monotonicity sandwiches the normalized mixing time between these two values; let $\delta \downarrow 0$.

For $0 < \varepsilon \leq e^{-1}$, every fixed $s < 1$ has limiting distance $e^{-s} > \varepsilon$, while every $s > 1$ has limiting distance zero. The same sandwich gives $t_d(\varepsilon)/T_d \to 1$, including equality $\varepsilon = e^{-1}$. This proves (6) without asserting a limit at the jump point $s = 1$.

The exact identity

$$\gamma_d T_d = \frac{1}{2} \log d + \beta_d T_d$$

and (11) prove the product condition. The thresholds $1/4$ and $3/4$ give the ratio in Theorem 1.1. Its limit is strictly less than one, proving absence of cutoff. The jump in the limiting profile at $s = 1$ is insufficient for cutoff, because an order-one amount of mixing has already occurred gradually on the same scale.

7 Relation to previous constructions and verification

Rare restarting is a classical obstruction to general product-condition criteria; see [3, 4]. We do not claim the projected identity (8) as a new phenomenon. Uniform nonzero cube generators alone give a reset rate too large relative to coordinate flips. The auxiliary group in (2) supplies enough distinct local generators to tune their relative rate while keeping the full graph unweighted. Lemma 3.1 then verifies that this auxiliary group does not change the limiting mixing profile. These are the construction's relevant additional ingredients.

Recent work on restart perturbations [5] concerns perturbations and stationary distributions of general reversible chains; it must not be confused with a prior proof of this particular unweighted family. Likewise, long-range walks on products of cyclic groups [6] provide spectral descriptions and examples, but the presence of a similar state space alone does not identify the same counterexample. The accompanying comparison records the precise parts of the literature inspected and the limits of the novelty search.

The degree restriction is substantive. Since $|G_d| = 2^d m_d$, one has

$$\frac{\log D_d}{\log |G_d|} \longrightarrow \frac{1}{2}.$$

Thus degrees grow roughly as the square root of the number of vertices, up to subpower factors. The construction supplies no counterexample to bounded-degree transitive-expander questions or statements about typical random generating sets. Those problems must retain their separate labels.

The handwritten proof above checks graph membership, the exact projected semigroup, the conditional auxiliary law, all spectral sectors, and the limiting-time brackets. The formal companion supplies proofs of the exact generator, gap and scaling algebra. It does not construct the graph or the Markov law in Lean, and it is not a complete formalization of Theorem 1.1. A separate verification record distinguishes successful component compilation from the still-unfinished full formal proof. No priority claim or journal-acceptance claim follows from compilation.

References

[1] J. Salez, *Modern aspects of Markov chains: entropy, curvature and the cutoff phenomenon*, 2025, Problem 2.4. arXiv:2508.21055.

[2] J. Hermon and S. Olesker-Taylor, *Cutoff for almost all random walks on Abelian groups*, J. Eur. Math. Soc. **28** (2026), 1913–1977. Section 7.4, Conjecture 1. doi:10.4171/JEMS/1743.

[3] D. A. Levin and Y. Peres, with E. L. Wilmer, *Markov Chains and Mixing Times*, second edition, American Mathematical Society, 2017. Chapter 18. Author's book page.

[4] J. Salez, *Cutoff for non-negatively curved Markov chains*, J. Eur. Math. Soc. **26** (2024), 4375–4392. doi:10.4171/JEMS/1348.

[5] D. Vial and V. Subramanian, *Restart perturbations for reversible Markov chains: trichotomy and pre-cutoff equivalence*, Random Structures & Algorithms (2025). doi:10.1002/rsa.70007.

[6] R. Griffiths and S. Mano, *A class of long range circulant random walks on* $\mathbb{Z}_q^d$, Stochastic Processes and their Applications **201** (2026), 105038. doi:10.1016/j.spa.2026.105038.